

Measuring Tree Canopy Height using SAR Interferometry

MTech Seminar

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by

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Abstract

Forest canopy height is an important measurement used to estimate biomass, calculate carbon stocks, study wildlife habitats, and monitor forest health. This report explains a step-by-step and easy-to-repeat workflow for estimating canopy height using Sentinel-1 SLC data. The processing is done mainly in ESA's SNAP software, with final corrections in QGIS. The workflow includes all major steps: applying orbit corrections, splitting and debursting TOPS data, accurate co-registration, Enhanced Spectral Diversity (ESD), creating the interferogram, applying Goldstein filtering, unwrapping the phase using SNAPHU, removing topographic phase using a DEM, and converting the phase into height to generate an interferometric DSM. The final canopy height model (CHM) is created by subtracting a ground DEM from this DSM.

When this workflow was applied to the IIT Dharwad forest area, the InSAR-based CHM showed much higher canopy heights (around 60–80 m) than what field data and other sources suggest (around 16–20 m). This report looks into why these values are overestimated and identifies possible reasons, such as phase unwrapping problems, errors in the DEM, issues in baseline estimation, and loss of coherence in dense vegetation when using C-band data. The report also suggests ways to reduce these errors, including using multiple interferograms (stacking), applying bias correction using ICESat-2 data, and combining C-band with L-band observations. All SNAP/SNAPHU parameters and QGIS steps are documented to ensure the workflow can be reproduced easily.

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1 Introduction

1.1 Motivation and significance

Tree canopy height plays a major role in understanding forests. It helps us estimate above-ground biomass, evaluate how suitable an area is for wildlife, measure how much carbon a forest can store, and track how forest structure changes over time. Traditional methods like field surveys and airborne LiDAR give very accurate height information, but they are costly, time-consuming, and usually cover only small areas.

Satellite remote sensing provides a practical alternative because it can capture data repeatedly and over large regions. Among different satellite technologies, Synthetic Aperture Radar (SAR) is especially useful. Unlike optical sensors, SAR does not rely on sunlight and can penetrate clouds, allowing data collection in all weather and at any time of day. SAR also records phase information, which can be used through interferometry (InSAR) to estimate height with high vertical sensitivity. This makes SAR-based InSAR a powerful tool for mapping canopy height across wide landscapes in a consistent and cost-effective way.

1.2 Goal of this work

The main aim of this study is to build and clearly document a complete Sentinel-1 InSAR workflow for generating canopy height models (CHMs) for the IIT Dharwad forest area. The results are compared with available reference data to check how accurate they are. The study also looks closely at where the method fails, why errors occur, and what practical steps can be taken to reduce them. Throughout the work, special attention is given to reproducibility by saving all metadata, orbit files, and by carefully noting every SNAP and QGIS step used in the process.

2 SAR and InSAR Interferometry

2.1 SAR Fundamentals

Synthetic Aperture Radar (SAR) is an **active microwave remote sensing system**, meaning it transmits its own radar pulses and records the echoes returning from the Earth’s surface. This allows SAR to acquire data **day or night** and under **all weather conditions**, including clouds and rain.

A SAR image is composed of **complex-valued pixels**, each containing two components: amplitude and phase. A complex SAR pixel can be written as

$$s(\mathbf{r}) = A(\mathbf{r}) e^{j\phi(\mathbf{r})},$$

where $A(\mathbf{r})$ is the signal amplitude and $\phi(\mathbf{r})$ is the phase (in radians).

- **Amplitude** mainly depends on surface roughness, moisture, and structural properties of the target. It forms the intensity image typically displayed in SAR products.
- **Phase** represents the portion of the radar wave cycle at the time it was received. It is directly related to the **range distance** between the sensor and the ground target, but only *modulo* the radar wavelength.

Because Sentinel-1 operates at **C-band** (wavelength ~ 5.6 cm), even very small path-length differences produce measurable phase changes. This makes SAR phase extremely sensitive to changes in height or ground movement.

The phase component is what enables **interferometric SAR (InSAR)**. By comparing the phase from two SAR acquisitions taken from slightly different satellite positions, tiny differences in range can be measured. These differences can then be converted into:

- surface elevation,
- canopy height,
- or ground deformation (e.g., subsidence, landslides, earthquakes).

Thus, while amplitude provides information on how the surface reflects radar energy, the

phase is the key quantity used to extract height information in this study.

2.2 Interferogram Formation

In interferometric SAR (InSAR), height or deformation information is extracted by comparing the phase of two complex Single Look Complex (SLC) images acquired from slightly different satellite positions. Let s_1 be the *master* image and s_2 be the *slave* image. Each pixel in these images contains amplitude and phase information.

The interferogram is formed by multiplying the master image with the complex conjugate of the slave:

$$I(\mathbf{r}) = s_1(\mathbf{r}) s_2^*(\mathbf{r}) = A_1(\mathbf{r}) A_2(\mathbf{r}) e^{j(\phi_1(\mathbf{r}) - \phi_2(\mathbf{r}))}.$$

This product removes the absolute phase and keeps only the **phase difference** between the two observations:

$$\Delta\phi(\mathbf{r}) = \phi_1(\mathbf{r}) - \phi_2(\mathbf{r}).$$

The resulting interferogram contains a wrapped phase image (values between $-\pi$ and $+\pi$) and an amplitude image. The phase difference $\Delta\phi$ is the key quantity used for estimating height or surface displacement. However, the measured phase is a mixture of several physical effects:

- **Topographic height:** The main component exploited for generating DEMs or canopy height models. A difference in look angle between the master and slave creates a phase shift proportional to terrain elevation.
- **Residual orbital errors:** Imperfect knowledge of satellite positions causes long-wavelength phase ramps across the interferogram.
- **Atmospheric delays:** Variations in the troposphere (humidity, pressure) and ionosphere (electron content) change the radar signal's travel speed, introducing additional phase shifts.
- **Surface deformation:** If the ground moved between the two acquisitions (e.g., subsidence, uplift, landslides), this displacement appears as an extra phase term.
- **Noise and decorrelation:** Thermal noise, temporal decorrelation (vegetation change), volume scattering, and geometric decorrelation reduce coherence and add random errors to the phase.

In practice, the goal of InSAR processing is to isolate the **topographic phase** from these other components. This requires careful co-registration, filtering, unwrapping, and removal of non-topographic terms before converting the phase into height.

2.3 Phase-to-Height Relationship

In InSAR, the height of the terrain (or canopy) is estimated by converting the measured topographic phase into elevation. Under small-baseline and near-bistatic geometry assumptions, the phase contribution due to topography can be written as:

$$\Delta\phi_{\text{topo}} = \frac{4\pi}{\lambda} \frac{B_{\perp} h}{r \sin \theta},$$

where:

λ radar wavelength,

$B_{\perp}$ perpendicular component of the spatial baseline,

h terrain or canopy height above a reference surface,

r slant range distance from satellite to ground,

θ incidence angle of the radar beam.

This equation shows that the topographic phase is directly proportional to the height of the surface. A larger perpendicular baseline produces a larger phase difference for the same height, making the interferogram more sensitive to vertical variations.

By rearranging the above equation, we obtain the **phase-to-height inversion** formula:

$$h = \frac{\lambda r \sin \theta}{4\pi B_{\perp}} \Delta\phi_{\text{topo}}.$$

This expression allows us to convert the unwrapped phase into actual height values to generate a digital surface model (DSM) or canopy height model (CHM).

Height–baseline trade-off: While a larger $B_{\perp}$ increases height sensitivity (because height produces a stronger phase response), it also increases *geometric decorrelation*, which reduces coherence and makes the interferogram noisier. Therefore, InSAR processing always involves a balance: choosing baselines large enough for good height resolution but small enough to maintain usable coherence.

2.4 Coherence and Decorrelation

Coherence (γ) is a key measure of the **similarity and phase stability** between two SAR acquisitions. It indicates how reliable the interferometric phase is. Mathematically, coherence is defined as:

$$\gamma = \frac{|\langle s_1 s_2^* \rangle|}{\sqrt{\langle |s_1|^2 \rangle \langle |s_2|^2 \rangle}},$$

where the angle brackets $\langle \cdot \rangle$ denote local spatial averaging.

- $\gamma \approx 1$: The two signals are highly correlated. Phase is stable and the interferogram

is of good quality.

- $\gamma < 0.3\text{--}0.4$: Coherence is low. Phase becomes noisy, and phase unwrapping often fails.

Sources of Decorrelation

A loss of coherence (decorrelation) occurs when the scattering properties of the surface change between the two acquisitions, or when the imaging geometry differs too much. Major sources include:

- **Temporal decorrelation:** Physical changes on the ground (e.g., vegetation growth, wind movement, moisture variation) reduce similarity between the two SLCs. Vegetated areas are especially prone to this.
- **Geometric decorrelation:** Large perpendicular baselines ($B_{\perp}$) change the viewing angle too much, causing different parts of the canopy or terrain to be illuminated.
- **Volume decorrelation:** In forests, the radar signal scatters multiple times inside the canopy volume. Because different layers of the forest (leaves, branches, trunks) contribute differently in each acquisition, the phase becomes inconsistent. This effect is particularly strong for C-band SAR.
- **Thermal noise:** Random noise in the radar system also reduces coherence.

In dense vegetation such as tropical forests, **volume decorrelation** is the dominant factor. As a result, coherence values in C-band Sentinel-1 data often drop significantly, especially when the temporal separation between acquisitions is long. Low coherence directly affects the reliability of phase unwrapping and therefore limits the accuracy of canopy height estimation in such regions.

3 Study Area and Datasets

3.1 IIT Dharwad forest AOI

The area of interest (AOI) is forested land inside and around the IIT Dharwad campus. Vegetation is mixed deciduous and planted, with patch sizes varying from small groves to contiguous forest stands.



Figure 3.1: IIT Dharwad forest.

3.2 Primary datasets

- **Sentinel-1 Level-1 SLC:** Master and slave acquisitions chosen for short temporal baseline 12 days. Files used:
S1A_IW_SLC__1SDV_20251029T005607_20251029T005635_061633_07B319_B413.SAFE (master),
S1A_IW_SLC__1SDV_20251017T005607_20251017T005635_061458_07AC11_93AF.SAFE (slave).

- **DEM:** SRTM 30 m (OpenTopography) used as topographic reference.
- **Ancillary:** Datasets used for interpretation and validation:
 - **Global Canopy Height Map (2020)** accessed through the Google Earth Engine application
<https://nlang.users.earthengine.app/view/global-canopy-height-2020>.

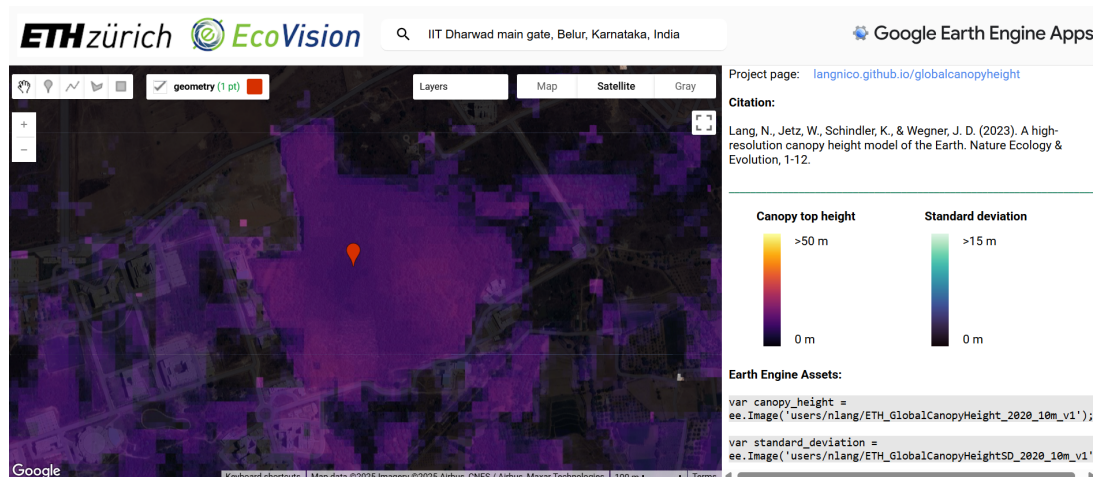


Figure 3.2: Reference canopy height map.

4 Processing Workflow: SNAP & QGIS

This chapter provides a complete, step-by-step description of the processing workflow implemented in ESA SNAP and QGIS, along with recommended parameter settings, explanations, and expected outputs at each stage.

4.1 Data (SLC, DEM)

This step involves collecting and preparing all the required datasets before starting InSAR processing. Proper preparation ensures that the later steps such as co-registration, interferogram formation, and phase unwrapping work correctly.

Inputs

- **Sentinel-1 Level-1 SLC** (master and slave) — provides the complex SAR data needed for interferometry.
- **DEM** (SRTM 30 m) — used for DEM-assisted co-registration, topographic phase removal, and terrain correction.
- **Precise Orbit Files (POEORB)** — replace the default predicted orbit to reduce phase ramps caused by orbital errors.

Importing Files into SNAP

1. Open SNAP → **File** → **Open Product**.
2. Select the master and slave **.SAFE** folders.
3. Import the DEM through **File** → **Import**.

SNAP automatically reads all metadata swath, bursts, Doppler, incidence angle.

Summary

In this stage, all raw datasets—SLC, DEM, and orbit files—are properly imported and corrected. This provides a clean, reliable starting point for co-registration and later InSAR processing.

4.2 Coregistration

Coregistration aligns the slave image to the master image so that both acquisitions match pixel-by-pixel. This step is critical because the interferogram is generated by subtracting the phase of the slave from the master. Any misalignment leads to loss of coherence and incorrect height estimates.

Master–Slave Selection

For this study:

- **Newer acquisition** → chosen as **Master**
- **Older acquisition** → used as **Slave**

The newer SLC is preferred as master because it usually has more accurate orbit information and metadata.

SNAP Operator

S-1 Back-Geocoding

Steps to Perform Coregistration in SNAP

1. Load both SLC products into SNAP.
2. Go to: Radar → Coregistration → S-1 TOPS Coregistration.
3. Select the **newer SLC as Master** and the **older SLC as Slave**.
4. Choose the correct **IW sub-swath** (IW1, IW2, or IW3).
5. Enable **DEM-assisted Back-Geocoding** (recommended for forests).
6. Keep default interpolation and resampling settings.
7. Run the operator to generate the co-registered slave image.

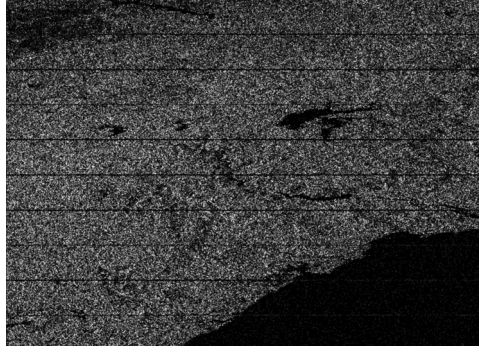


Figure 4.1: Co-registered SLC.

4.3 Interferogram Formation

Interferogram formation is the central step of InSAR processing. Once the slave image has been accurately co-registered to the master, the complex interferogram is created by multiplying the master SLC with the complex conjugate of the slave. The resulting phase difference represents the wrapped interferometric phase.

SNAP Operator

Radar → Interferometric → Products → Interferogram Formation

Steps to Generate the Interferogram in SNAP

1. Load the **co-registered master and slave** product (output from Coregistration).
2. Go to Radar → Interferometric → Products → Interferogram Formation.
3. Select the master and slave bands (VV–VV).
4. Enable the following options:
 - **Subtract Flat Earth Phase** (recommended)
 - **Include Coherence Estimate**
5. Choose a suitable coherence window (e.g., 5×5).
6. Keep default multilook factors at this stage (multilooking happens later).
7. Run the operator to generate the interferogram.

Outputs

- **Wrapped interferogram** — phase values between $-\pi$ and $+\pi$ representing the phase difference between master and slave acquisitions.

- **Coherence map** — shows pixel-wise phase stability.

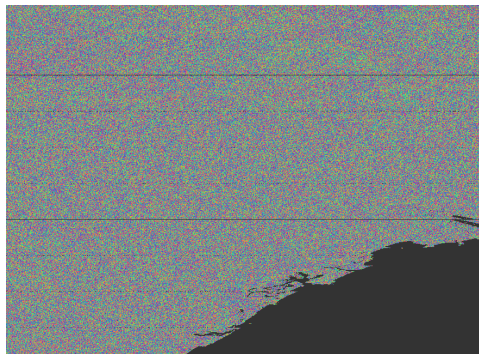


Figure 4.2: Wrapped interferogram.

Importance of Coherence

Coherence is a critical quality measure:

- High coherence ($\gamma > 0.4$) $\rightarrow$ reliable phase, suitable for unwrapping.
- Low coherence ($\gamma < 0.3$) $\rightarrow$ noisy or decorrelated areas (dense forest, water, temporal change).

These outputs determine whether the interferogram is suitable for phase unwrapping and height estimation in subsequent steps.

4.4 Deburst (TOPS Mode)

Sentinel-1 IW (Interferometric Wide Swath) data is acquired in **TOPS mode**, where each subswath (IW1, IW2, IW3) is divided into several bursts. Each burst is recorded with a different azimuth steering angle. As a result, the master and slave SLCs contain burst-wise phase discontinuities that must be removed before generating a continuous interferogram.

Why Debursting is Required

- TOPS azimuth steering causes phase jumps at burst boundaries.
- These discontinuities appear as vertical lines or stripes in the wrapped interferogram.
- If not removed, phase unwrapping will fail and produce height errors.

Therefore, debursting is a mandatory step in Sentinel-1 TOPS InSAR processing.

SNAP Operator

Radar $\rightarrow$ Sentinel-1 TOPS $\rightarrow$ TOPS Deburst

Steps to Perform Debursting in SNAP

1. Open the **co-registered** product (output from Coregistration).
2. Go to: Radar → Sentinel-1 TOPS → TOPS Deburst.
3. Select the co-registered master–slave product.
4. Ensure the correct IW subswath (IW1/IW2/IW3) is selected.
5. Keep default settings (SNAP automatically handles burst stitching).
6. Run the operator to produce a single continuous, burst-free image.

Purpose

- Merges all bursts into one seamless SLC image.
- Removes azimuth phase jumps across burst boundaries.
- Ensures phase continuity, which is essential for:
 - stable coherence calculation,
 - smooth interferogram formation,
 - successful phase unwrapping.

After debursting, the interferogram becomes spatially continuous and ready for the filtering and unwrapping stages.

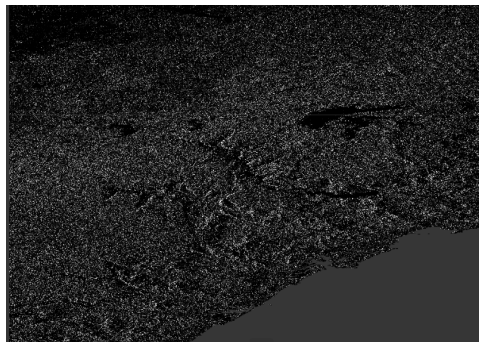


Figure 4.3: Intensity after Deburst.

4.5 Phase Filtering (Goldstein Filtering)

Phase filtering is applied to reduce high-frequency phase noise in the interferogram while preserving the underlying fringe pattern. This step significantly improves the quality of the unwrapped phase, especially in vegetated or low-coherence regions where phase residues are common.

Why Filtering is Needed

- The wrapped interferogram contains speckle and high-frequency noise.
- Noise increases phase residues, which makes phase unwrapping unstable.
- Filtering enhances coherence and smooths phase variations without destroying fringes.

Goldstein filtering is preferred because it adaptively suppresses noise based on spectral characteristics of the interferogram.

SNAP Operator

Radar → Interferometric → Filtering → Goldstein Phase Filter

Steps to Apply Goldstein Filtering in SNAP

1. Load the wrapped interferogram (output from “Interferogram Formation” or “De-burst” depending on workflow).
2. Go to: Radar → Interferometric → Filtering → Goldstein Phase Filter.
3. Select the interferogram phase band.
4. Set the filtering parameters:
 - **Alpha (filter strength):** 0.6–0.8 (higher = stronger filtering)
 - **Window size:** 32–64 pixels depending on noise level
5. Keep default multilooking settings if multilook was not performed earlier.
6. Run the operator to generate the filtered interferogram.

Recommended Parameters

- **Filtering strength:** $\alpha = 0.6\text{--}0.8$ (Balances noise suppression and fringe preservation.)
- **Window size:** 32–64 (Larger windows provide stronger smoothing but may blur narrow fringes.)

Outcome

Filtering:

- increases coherence,
- reduces phase jumps,

- lowers the number of residues,
- and stabilizes the phase for unwrapping.

A clean, filtered interferogram is essential before exporting to SNAPHU for phase unwrapping.

4.6 Phase Unwrapping (SNAPHU)

Phase unwrapping converts the wrapped phase (range $-\pi$ to π) into a continuous phase field. This step is essential because the phase-to-height conversion requires unwrapped phase values. Sentinel-1 TOPS data, especially over vegetation, often contains residues and discontinuities; therefore, a robust external unwrapping tool like SNAPHU is used.

Exporting Data from SNAP to SNAPHU

SNAP provides a dedicated exporter that prepares all required files.

1. Select the filtered interferogram product.
2. Go to: **Radar** → **Interferometric** → **Unwrapping** → **Snaphu Export**.
3. Choose the interferogram phase band.
4. Enable **Export Coherence** (used as a quality mask).
5. Select **TOPO** mode for height estimation.
6. Set the tile size (e.g., 2048) for memory-efficient processing.
7. Export to an output directory (SNAPHU-ready folder).

This step generates:

- Wrapped interferogram (**.img**)
- Coherence map (**.img**)
- Configuration file (**snaphu.conf**)
- Supporting metadata files

Running SNAPHU from the Command Line

SNAPHU must be run externally because SNAP does not unwrap the phase internally.

Typical SNAPHU Command

Inside the exported folder, run:

Where:

- `wrapped_phase.img` — exported phase file
- `width` — number of columns (given in metadata file)
- `snaphu.conf` — configuration file generated by SNAP

Depending on the scene size, unwrapping may take a few minutes to more than 30 minutes.

Typical SNAPHU Settings

- Cost mode: `DELAUNAY` (recommended for topography)
- Init method: `QUAD_TREE`
- Residue cost: 40
- Tile size: 2048

These settings help handle noisy forest areas by weighting coherence appropriately.

Importing Unwrapped Phase Back into SNAP

Once SNAPHU produces the unwrapped phase file:

1. Go to: `Radar` → `Interferometric` → `Unwrapping` → `Snaphu Import`.
2. Select the original exported folder.
3. Choose the `unwrappedPhase.img` generated by SNAPHU.
4. SNAP attaches the unwrapped phase band to the original interferogram product.

Outcome

The final output is:

- A continuous unwrapped phase band (no $-\pi$ to π jumps)

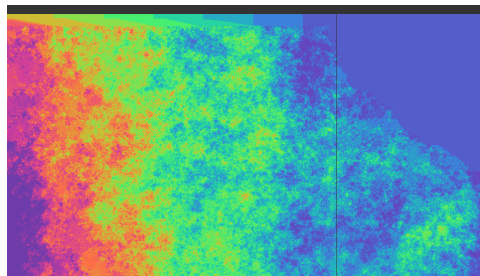


Figure 4.4: Unwrapped Phase.

4.7 Phase-to-Height

Once the phase has been successfully unwrapped, the next step is to convert the unwrapped phase into physical elevation values. This process transforms the interferometric phase differences into a Digital Surface Model (DSM), representing the height of the canopy or ground surface in radar geometry.

Concept

The unwrapped phase contains all remaining height-related information after removing flat-earth and topographic components. Using the imaging geometry, baseline, and radar wavelength, this phase can be converted to elevation.

Formula

The phase-to-height conversion is given by:

$$h = \frac{\lambda r \sin \theta}{4\pi B_{\perp}} \cdot \phi_{\text{residual}}$$

Where:

- λ — radar wavelength (Sentinel-1: 5.6 cm)
- r — slant range distance from satellite to target
- θ — incidence angle
- $B_{\perp}$ — perpendicular baseline
- ϕ_{residual} — phase after removing DEM topographic phase

A small error in baseline or angle can directly scale the final height, which is why accurate processing in earlier steps is critical.

Procedure in SNAP

1. Load the interferogram containing the unwrapped phase.
2. Go to **Radar** → **Interferometric** → **Products** → **Phase to Elevation**.
3. Select the unwrapped phase band.
4. Provide:
 - the same DEM used earlier,
 - incidence angle band,

- and baseline information (SNAP reads it automatically).
5. SNAP internally computes the DEM phase and subtracts it from the unwrapped phase.
 6. The remaining residual phase is converted into height using the baseline geometry.

Outputs

- **Radar-coded DSM (slant-range geometry)** — represents canopy or surface height relative to the sensor geometry.

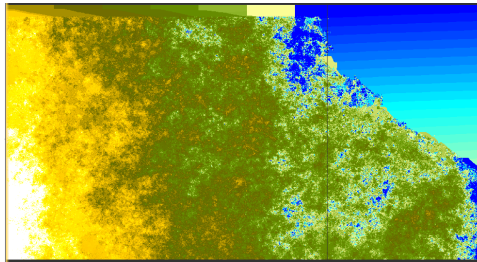


Figure 4.5: Radar coded DSM.

This DSM still needs to be geocoded into map coordinates using Terrain Correction in the next step.

4.8 Terrain Correction

The DSM obtained from phase-to-height conversion is still in radar geometry (slant-range coordinates). Terrain Correction converts this radar-coded DSM into a map-projected product (e.g., UTM), so that each pixel corresponds to a real geographic location on the ground. This step also corrects geometric distortions such as foreshortening and terrain-induced shifts.

SNAP Operator

Radar → Geometric → Terrain Correction → Range-Doppler Terrain Correction

Purpose

- Convert slant-range DSM into a map-projected DSM.
- Use the DEM to correctly position pixels on the ground.
- Apply geoid correction to convert ellipsoidal heights to orthometric heights (optional but recommended).

- Remove geometric distortions caused by viewing angle and terrain.

Steps in SNAP

1. Load the radar-coded DSM produced from the “Phase-to-Height” step.
2. Go to:
Radar → Geometric → Terrain Correction → Range-Doppler Terrain Correction.
3. Select the correct DEM (same one used previously).
4. Choose the map projection (UTM zone corresponding to your study area).
5. Set the output pixel size (e.g., 10 m or 30 m depending on analysis needs).
6. Select resampling method: Bilinear or Cubic for smoother DSM.
7. Run the operator to produce a geocoded height map.

Output

- **Geocoded DSM (GeoTIFF)** — elevation mapped to UTM coordinates and ready for export to QGIS.

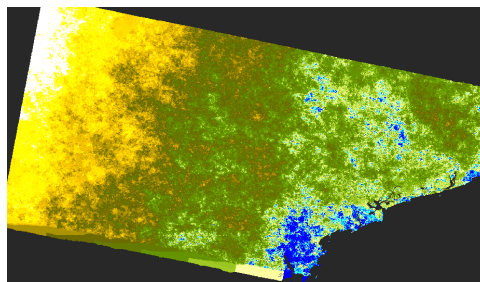


Figure 4.6: Geocoded DSM.

4.9 Export DSM

After terrain correction, the DSM is properly geocoded and ready to be transferred to GIS software for canopy height computation. This step prepares the DSM in a standard GeoTIFF format that can be loaded directly into QGIS.

Purpose

To export the geocoded DSM from SNAP in a consistent projection and resolution so that it can be used for raster calculations ($\text{DSM} - \text{DEM}$) inside QGIS.

Procedure in SNAP

1. Open the terrain-corrected DSM product.
2. Go to:
File → Export → GeoTIFF.
3. Choose the output file name and destination folder.
4. Make sure the following settings are correct:
 - **CRS:** Same UTM projection used during terrain correction.
 - **Pixel size:** Should match DEM resolution (10 m or 30 m).
 - **Band name:** Ensure the elevation band is clearly labeled (e.g., “height” or “DSM”).
5. Confirm that no unintended scale factors are applied in the metadata.
6. Export the file as a standard `.tif` raster.

Output

- **Geocoded DSM (GeoTIFF)** — ready for use in QGIS for CHM generation.

5 QGIS Post-processing and CHM Generation

Once the DSM has been terrain-corrected and exported from SNAP, the remaining processing is performed in QGIS. The goal is to prepare the DSM and DEM on a common grid and generate the final Canopy Height Model (CHM) by simply subtracting ground elevation from surface elevation.

5.1 Preprocessing Steps in QGIS

Before computing the CHM, both input rasters (DSM and DEM) must be aligned and standardized. This ensures that each pixel in the DSM corresponds to the same ground location in the DEM.

1. **Load DSM and DEM into QGIS** Import the GeoTIFF DSM exported from SNAP and the DEM (SRTM or any ground DEM). Both rasters should appear in the Layers panel.
2. **Check Coordinate Reference System (CRS)** Ensure that both rasters use the same map projection (e.g., UTM). If not:
 - Right-click raster → **Export** → **Save As**
 - Reproject to the correct CRS

Units must be in meters for correct height difference calculations.

3. **Align Rasters** DSM and DEM must have:
 - identical pixel size,
 - identical extent,
 - identical grid alignment.

Use:

Raster → **Miscellaneous** → **Align Rasters** (GDAL warp tool)

- Choose DSM as the reference raster.
 - Set DEM as the input to be aligned.
 - Select resampling method: Bilinear or Cubic.
4. **Clip Rasters to AOI** To speed up processing and avoid unnecessary calculations:
- Load AOI boundary (shapefile or polygon).
 - Use **Raster → Extraction → Clip Raster by Mask Layer**.

This produces clipped DSM and DEM focused only on the forest study area.

5.2 Compute CHM (Raster Calculator)

Once both the DSM and DEM are aligned to the same grid and clipped to the AOI, the Canopy Height Model (CHM) can be generated. The CHM represents the height of the vegetation canopy above the ground and is obtained by subtracting the ground elevation (DEM) from the surface elevation (DSM).

Raster Calculator Expression

Open the QGIS Raster Calculator and use:

$$\text{CHM} = \text{DSM@1} - \text{DEM@1}$$

Here:

- **DSM@1** refers to the first band of the DSM raster (surface height)
- **DEM@1** refers to the first band of the DEM raster (ground height)

This produces a new raster where each pixel value represents canopy height in meters.

Post-processing Steps

To refine the CHM and remove artefacts:

- **Remove negative heights** Due to minor DEM–DSM misalignments or noise, some pixels may produce negative values. Replace them with:
 - 0 meters (if small errors), or
 - NoData (if large anomalies).

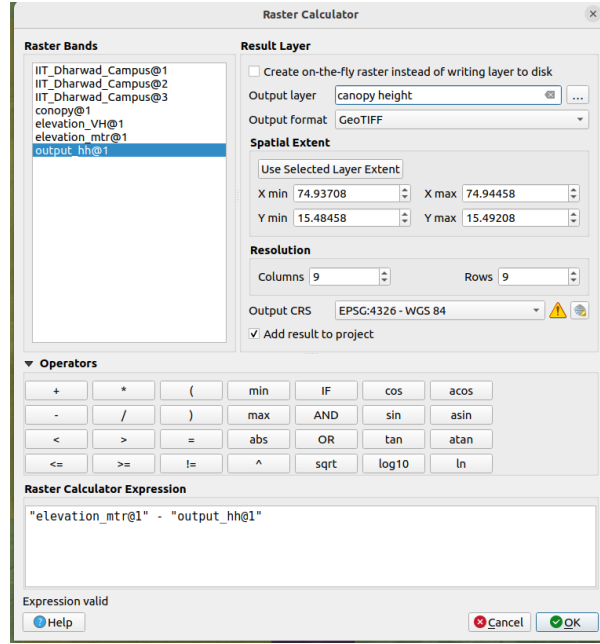


Figure 5.1: QGIS Raster Calculator used to compute CHM.

5.3 Validation and Statistics

Validation helps assess how reliable the InSAR-derived CHM is compared to real forest height conditions.

5.3.1 Zonal Statistics

Using the QGIS Zonal Statistics tool, basic metrics such as mean, median, standard deviation, and minimum–maximum heights can be computed over selected forest patches or AOI polygons. These statistics reveal height distribution and highlight any abnormal values.

Reference Data Comparison

The CHM can be compared against:

- Global canopy height maps such as GCH 2020.

Overlaying these datasets helps identify systematic height bias or areas affected by low coherence and unwrapping errors.

Example

For the IIT Dharwad forest:

- Expected canopy height: **16–20 m**

- InSAR-derived CHM: **60–80 m**

This mismatch led to further diagnostic analysis discussed in the next chapter.

6 Results and Diagnostics

The key outputs of the InSAR workflow are the geocoded Digital Surface Model (DSM) and the final Canopy Height Model (CHM). These visual products help assess whether the processing steps—interferogram formation, unwrapping, phase-to-height conversion, and terrain correction—performed correctly.

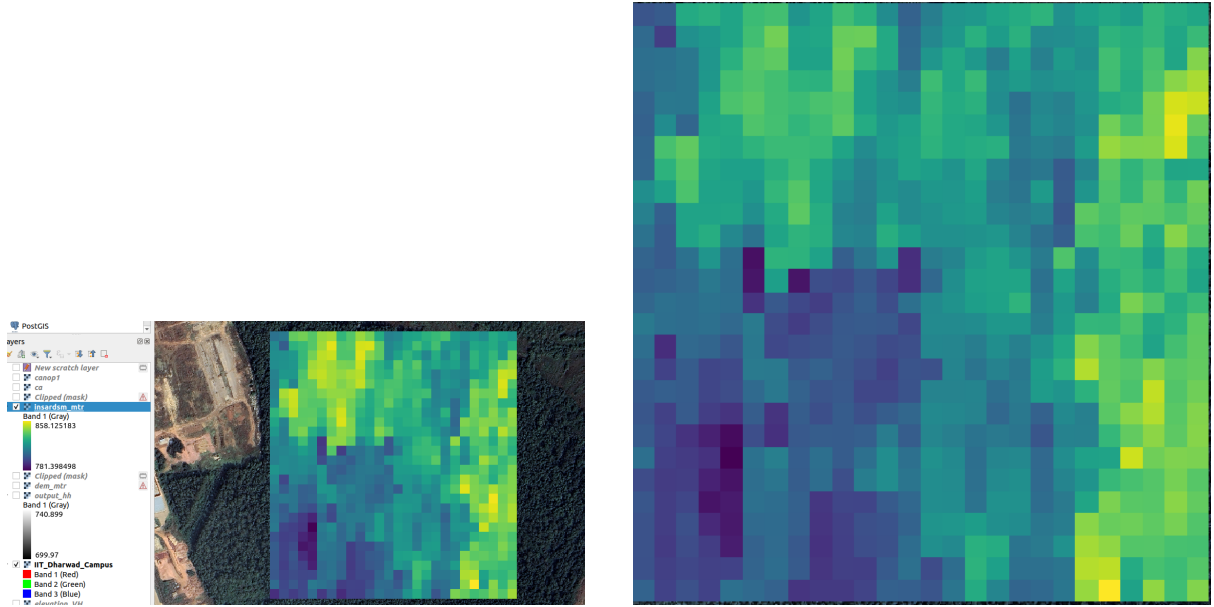


Figure 6.1: Left: InSAR-derived DSM. Right: CHM generated as $\text{DSM} - \text{DEM}$.

The DSM (left) shows the elevation of all surface features as captured by the radar, including tree canopy, built structures, and terrain. Smooth transitions indicate successful phase unwrapping, whereas sharp jumps or streaks may indicate unwrapping errors or low-coherence areas.

The CHM (right) highlights only the vegetation height by removing ground elevation using a DEM. Areas with dense forest appear as higher values, while open land or cleared regions show near-zero height. Visual inspection of the CHM helps quickly identify:

- regions where canopy height appears unusually high or low,
- artefacts caused by unwrapping discontinuities,

- gaps due to low coherence ,
- overall spatial patterns in vegetation height.

These visual maps provide the foundation for the quantitative validation and diagnostic analysis presented in the next sections.

6.1 Observed Statistics

Basic height statistics were computed for the IIT Dharwad forest area to compare the InSAR-derived canopy heights with expected values from reference datasets. The results indicate a clear mismatch between the two, suggesting systematic bias in the InSAR workflow.

Region	Expected (m)	Obtained (m)	Notes
IIT Dharwad AOI	16–20	60–80	Strong overestimation

Table 6.1: Summary of canopy height comparison.

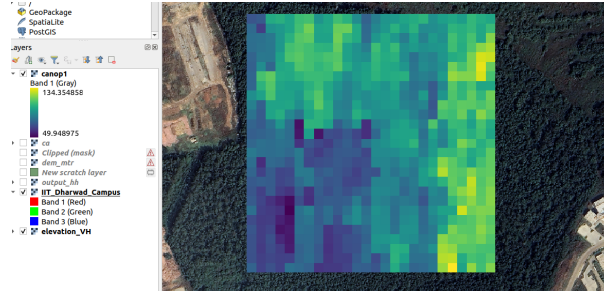


Figure 6.2: Canopy Height.

The obtained height range (60–80 m) is significantly higher than the expected canopy heights (16–20 m) reported by reference products such as GEDI/ICESat-2 and GCH 2020. This discrepancy suggests the presence of processing issues such as DEM bias, unwrapping errors, incorrect baseline scaling, or decorrelation effects, which are examined in detail in the diagnostic section that follows.

6.2 Diagnostic checklist for overestimation

When CHM shows excessive heights, check the following items:

1. **DEM bias:** SRTM or other DEMs may include vegetation or have vertical bias. Verify DEM source and, if possible, use higher-quality ground DEM or correct using GCPs.

2. **Unwrapping failures:** Residues and branch cuts in SNAPHU can cause large localized height jumps. Visual inspection of the unwrapped phase is essential.
3. **Baseline errors:** Verify the perpendicular baseline $B_{\perp}$ used for phase-to-height scaling. A scale error in $B_{\perp}$ directly scales heights.
4. **Co-registration:** Even small misregistrations produce phase noise and can bias heights.
5. **Volume scattering and phase center bias:** In vegetated areas, the radar scattering center is inside the canopy volume, not at the canopy top, causing bias in retrieved heights (C-band is sensitive to mid-canopy scattering).
6. **Atmospheric effects:** Tropospheric delays produce low-frequency phase ramps that can bias height if not corrected.

Each of these items should be systematically investigated when encountering large discrepancies between InSAR-derived CHM and reference heights.

7 Recommendations and Future Work

7.1 Immediate Mitigations

Several practical steps can be taken to reduce the height overestimation observed in the CHM and to improve the reliability of the InSAR-derived DSM.

- **Use multiple interferometric pairs (stacking or SBAS).** Combining several interferograms reduces random noise, averages out atmospheric effects, and stabilizes the phase. A stacked DSM typically shows much lower variance and fewer unwrapping artefacts compared to a single-pair result.
- **Apply bias correction using ICESat-2 or GEDI LiDAR points.** Spaceborne LiDAR provides accurate canopy-top heights. Comparing these points with the InSAR DSM allows:
 - estimation of a constant or spatially varying bias,
 - correction of the DSM before CHM calculation.
- **Mask low-coherence regions.** Restrict CHM computation to pixels with coherence above a threshold (e.g., $\gamma > 0.4$). This prevents unreliable or decorrelated pixels—common in dense vegetation and water bodies—from influencing height results.
- **Re-run SNAPHU with adjusted parameters.** Changing initialization (e.g., MST, QUAD_TREE) or cost modes can resolve unwrapping errors. Comparing outputs from multiple SNAPHU runs helps identify:
 - phase discontinuities,
 - large unwrapping jumps,
 - areas affected by noise or residues.

7.2 Longer-term Improvements

Beyond immediate fixes, several long-term strategies can significantly improve the accuracy and robustness of canopy height estimation using InSAR.

- **Use L-band SAR (e.g., ALOS-2).** L-band signals penetrate deeper into forest canopies and experience less volume decorrelation than C-band. This often yields height estimates closer to true canopy-top height and provides more stable interferograms in dense vegetation.
- **Fuse InSAR DSM with optical structure metrics.** Optical data from Sentinel-2 (NDVI, texture, LAI) can be combined with InSAR heights using regression or machine learning models. Such fusion approaches help correct systematic height biases caused by phase-center shifts in radar data.
- **Apply polarimetric decomposition.** Using dual-pol or full-pol SAR enables decomposition into surface, double-bounce, and volume scattering components. Understanding the scattering behavior helps interpret canopy structure and can support height corrections.
- **Perform atmospheric correction.** Tropospheric delays introduce large-scale phase ramps. Using external datasets like ERA5 reanalysis or GNSS-derived water vapor fields can help remove atmospheric artefacts and improve height consistency.
- **Improve validation with high-quality ground truth.** Field plots, terrestrial laser scanning (TLS), and airborne LiDAR provide accurate canopy height measurements. These datasets are essential for calibrating models, validating InSAR outputs, and quantifying uncertainties.

8 Practical Parameters and Example SNAP/SNAPHU Settings

8.1 Suggested SNAP Parameter Table

Table 8.1 provides a quick reference for the recommended operator settings used throughout the workflow. These values serve as a starting point and can be adjusted based on scene characteristics such as vegetation density, baseline length, and coherence levels.

Operation	Recommended values / notes
Apply Orbit	Use precise (POEORB) files; confirm correct timestamp match.
TOPS Split	Select correct IW sub-swath (IW1/IW2/IW3) covering AOI.
Deburst	Mandatory for TOPS SLCs; ensures phase continuity across bursts.
Co-registration	Achieve sub-pixel accuracy (≤ 0.1 pixel) in both range/azimuth.
Multilook	Typical settings: 4 (range) $\times 1$ or 4×2 (azimuth), depending on noise.
Goldstein filter	$\alpha = 0.6\text{--}0.8$; window size tuned to noise level (32–64).
SNAPHU mode	Use TOPO mode; cost = DELAUNAY or MCF; provide coherence map.
Phase-to-height	Allow SNAP to read geometry; verify $B_{\perp}$ in processing log.
Terrain correction	Output CRS in UTM; pixel size 10–30 m; enable EGM96 correction.

Table 8.1: Recommended SNAP parameter settings (adjust as needed for AOI).

8.2 Example SNAPHU Command (Conceptual)

SNAPHU is typically run externally via command line after exporting the interferogram and coherence map from SNAP. Below is a representative example of the command structure:

```
snaphu wrapped_phase.img width \
      -c snaphu.conf \
      --init-method=QUAD_TREE \
      --cost-mode=DELAUNAY \
      --resid-cost=40 \
      --tile-size=2048 \
      --logfile=snaphu.log
```

Notes

- `wrapped_phase.img` and `snaphu.conf` are generated by SNAP during the **SNAPHU Export** step.
- `width` (number of columns) is provided in the accompanying metadata file.
- Parameters such as tile size, cost mode, and initialization method should be adapted based on scene complexity, available memory, and coherence conditions.
- After unwrapping, the output file (`unwrappedPhase.img`) is imported back into SNAP using the **SNAPHU Import** tool.

A Appendix A: Useful Commands and Reproducible Steps

A.1 SNAP Graph Builder Tips

Using SNAP's Graph Processing Framework (GPF) helps ensure that the workflow is fully reproducible and can be run multiple times with consistent settings.

1. Build a processing graph containing the following sequence:

Read → Apply Orbit → TOPS Split → Deburst → Back-Geocoding → Interferogram Formation

2. Save the graph as an `.xml` file using: **File** → **Save Graph**.
3. Maintain versioned copies (e.g., `graph_v1.xml`, `graph_v2.xml`) whenever parameters change. This makes it easy to trace how each run was produced.
4. Use `gpt` (SNAP command-line tool) to execute the graph in batch mode for automation:

```
gpt insar_graph.xml -Pinput=<SLC folder> -Poutput=<output folder>
```

A.2 QGIS Reproducible Steps (Brief)

The following steps summarize a minimal but complete workflow to generate the CHM in QGIS.

1. Load the exported DSM and the DEM into QGIS.
2. Use **Raster** → **Projections** → **Warp (Reproject)** to ensure both rasters have the same CRS (e.g., UTM) and units in meters.
3. Align both rasters to the same grid using: **Raster** → **Miscellaneous** → **Align Rasters**. Select the DSM as the reference raster to ensure consistent resolution and

extent.

4. Compute the CHM using the Raster Calculator:

$$\text{CHM} = \text{DSM@1} - \text{DEM@1}$$

5. Use **Zonal Statistics** to compute mean or median canopy height for selected forest patches or AOI polygons.

A.3 Example BandMath Expression (SNAP)

This expression can be used within SNAP's BandMath tool to compute canopy height directly if DSM and DEM are already aligned:

$$\text{Canopy} = \text{DSM} - \text{DEM}$$

This produces a canopy height band inside SNAP, but final refinement and masking are recommended in QGIS.

B Appendix B: Checklist for Debugging Large CHM Errors

1. **Orbit files:** Confirm that `Apply-Orbit-File` used the correct **Precise (POE-ORB)** orbit file.
2. **Debursting:** Check that TOPS debursting is complete and no burst boundaries or vertical stripes remain in the interferogram.
3. **Co-registration:** Verify range/azimuth offsets are well below 0.1 pixel. Re-run with DEM-assisted options if needed.
4. **Coherence mask:** Inspect coherence map and apply a threshold (e.g., $\gamma > 0.35$ – 0.40) before unwrapping.
5. **Filtering and residues:** Ensure Goldstein filtering reduced noise and that residues are not clustering excessively.
6. **SNAPHU tests:** Try different `init-methods` (MST / QUAD_TREE) and `cost-modes` (DELAUNAY / MCF) to identify consistent unwrapped results.
7. **Unwrapped phase check:** Look for phase jumps, ramps, or tile-like discontinuities that indicate unwrapping failure.
8. **Baseline geometry:** Verify the $B_{\perp}$, incidence angle, and slant range values in SNAP logs. Incorrect geometry directly scales height.
9. **DEM quality:** Confirm DEM does not contain vegetation bias. Cross-check with alternative DEM sources if available.
10. **Reference comparison:** Compare DSM/CHM with GEDI, ICESat-2, or GCH2020 to identify systematic over- or underestimation.
11. **Bias correction:** If a consistent offset exists, compute a per-patch correction using LiDAR/GEDI reference points.

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